

As a **human computer interaction (HCI)** and **social computing** researcher, my work focuses on *understanding and reshaping how disruptive technologies (e.g. AI-mediated labor markets) interact with and affect the present and futures of workers*. To promote more collective, safe and prosperous worker communities, I draw on quantitative empirical methods (e.g. statistical analysis, LLM probes), grounded qualitative approaches (e.g. multi-stakeholder co-design/interviews in solidarity with workers) and system development (e.g. technological probes/interventions).

One type of worker I study is **platform-based gig workers**, whose labor and services are increasingly woven into the fabric of everyday life. For consumers, gigs offer convenient, on-demand help with everyday tasks (from Ubering, to caretaking, to creative and professional projects) by drawing on the wide-ranging expertise of a global workforce. For workers, the temporary and task-based nature of gigs represents *independence* from traditional jobs and organizations – affording them the perceived freedom to choose not only *when* and *where* to labor¹, but also *what* to work on. However, as platforms mediate between consumers and workers, they capitalize on the absence of legal protections for independent contractors to exploit gig workers physically, financially and psychologically [b]. With rising rates of worker turnover and displacement, how do we create transformative interventions that propel workforces toward more natural, collective and liberated future interactions? Overall, the primary themes of my research:

- ① employs large-scale computational methods to identify worker practices [1] and intentions [11]
- ② co-designs with multi-stakeholder communities to envision collective, sustainable futures [2, 5, 6]
- ③ develops computational [7] interventions to facilitate more joyful [10], pluralistic [3, 9] and natural user interactions.

At the intersection of the HCI, CSCW² and design communities, my thesis work applied human-centered techniques to understand *interactions between disruptive technologies (i.e. gig platforms) and stakeholders (i.e. workers, consumers, regulators)*, followed by the **design and development of alternative interventions** to probe and reshape oppressive technology-mediated relations. I contributed to 12 publications/workshop (7 as first-author, 2 as advising author) across a range of venues: ACM CHI, CSCW, UIST, DIS, CHIWORK and IEEE VL/HCC. Two of my first-authored publications received best paper and honorable mention recognitions at CHI and CHIWORK, and I acquired over \$224,000 in funding as a NSF GRFP recipient and a Block Center for Technology and Society grant awardee.

By investigating timely challenges through multi-stakeholder and multi-scalar approaches, I aim to continue *surfacing timeless, generalizable designs and interventions* that guide future AI-mediated workforces in navigating disruptive technological change. I am passionate about creatively applying my empirical and design expertise to promote futures of work that (1) evolve individual human capabilities (2) motivate collective and supportive social relations and (3) cultivate deliberative, trusting and pluralistic online communities.

① Understanding Worker Practices & Rationales via Large-Scale Data Analysis [1, 11]

Gig platforms commonly leverage a consumer-first rhetoric to justify and avoid accountability for unfair labor practices. But despite highly competitive, isolating platform environments, workers continue to willingly engage – suggesting the presence of underexamined intrinsic motivations that drive them to develop strategies for handling unexpected challenges. In my first pair of contributions, I present large-scale analyses that uncover (1) *how* freelancers communicate on platforms to achieve productivity and success (Fig. 1) as well as (2) crowdworker rationales on *why* they perform platform-mediated gigs despite harsh working conditions.

¹ For example, Uber marketed work flexibility through the slogan: “Drive when you want. Earn what you need.”

² Computer-supported cooperative work

To **scalably understand individual worker strategies**, we engaged in multi-month negotiations with collaborators to access a large-scale dataset (> 2 million messages, >58k freelancers, >56k projects) from a leading online freelancing platform. Applying regression analyses on these platform-mediated freelancer-client communications, I found statistically significant correlations between (1) personalized message content and higher *job acquisition rates* (2) standardized messaging schedules and increased *completion rates*, and (3) repeated uses of templates across projects that associated with greater bid volumes and *long term revenue* [1].

But *why* are workers motivated to continue engaging in online platform labor despite subpar conditions? We hypothesized that most workers' underlying beliefs about platforms remained undisclosed, under-examined, and likely misguided by platform-constructed illusions of flexibility. To explore latent worker *grievances* and *rationales*, we leveraged the persuasive power of language models to dissuade them from believing five common myths around gig work conditions. We collected workers' rationales for misguided beliefs and generated persuasive LLM responses to further probe their underlying incentives, revealing how **crowdworkers labor for platforms out of desires to preserve independence, autonomy and flexibility** [11].

② Co-designing Multi-Stakeholder Interventions for Platform Accountability [2, 5, 6, 7]

Mobilizing the data and efforts of online gig workforces revealed crucial strategies and rationales. But offline gig work requires support from higher-power stakeholders to enact infrastructural reforms that advance gig labor rights and protections.

To synergistically integrate priorities from multiple stakeholders, I led co-design workshops with policy experts, platform employees and workers to tackle five representative problems in gig work [2]. Our solution-oriented version of the speed-dating design method led stakeholders to generate and rank infrastructural interventions they are most prepared or motivated to implement (Fig. 2), which include visions of *stronger regulation*, *worker-owned coops* and *platform-government co-regulation*. Stakeholders also envisioned more incremental advances: platform-driven services/programs and (collectivist) technologies to enhance scheduling, surface resources and support financial well-being.

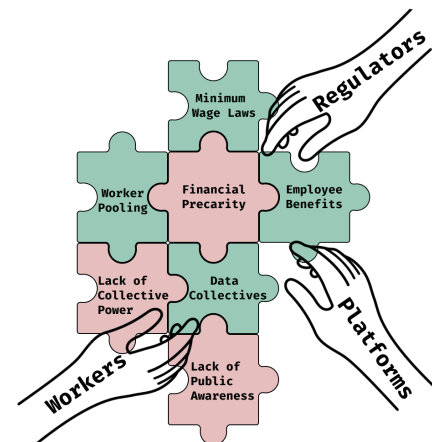


Fig. 2 Multi-Stakeholder Co-design of Solutions

Uncovering collective visions for infrastructural and regulatory change is a critical first step, but to achieve these in practice? In a CSCW '24 workshop, I led like-minded scholars to explore *worker data collectives* [6] for enacting counterdata production as well as advocacy and policy advances. I also interviewed policy experts from varying levels of office to surface prioritized initiatives for data-sharing (*i.e.* safety, fair pay) and design recommendations [5] that subsequently informed early-stage designs of [Gig2Gether](#) (Fig. 3), an open-sourced, cross-platform (pink-, white- and blue-collar domains) web app allowing users to track and share gig work data, view aggregate data visualizations and post qualitative stories about labor conditions that promote evidence-based advocacy and policymaking [6]. In a 7-day field evaluation with 16 participants, **Gig2Gether facilitated cross-platform mutual support** and financial reflection/planning, in addition to revealing latent worker desires for future use cases — including **collective algorithmic auditing and policy advocacy**. For one Gig2Gether user, the tool represented “*a door between [gig workers] and politicians*”.

③ Building Social Solidarity & Pluralistic Interactions amidst Evolving Tech [3, 10]

Alongside technological interventions [2], regulators in our workshops also called for increased and individualized policy. In response, I synthesized legal and policy documents, labor regulations, exemplary state bills and HCI scholarship to envision advances. In an award-winning position paper [3], I advocated for targeted and pluralistic policies that consider diverse task domains and intersectional worker backgrounds. But

while policymakers call for stronger labor regulations, platforms continuously rely on workers' non-employee status to evade employer accountabilities – causing them to shoulder *vulnerabilities like immaterial labor and emotional burdens, alone* [c]. Meanwhile, HCI research has overlooked the role of consumers in worker advocacy, which harms workers when uninformed consumers hold unrealistic expectations of service quality.

How to design interventions to **motivate consumer-worker solidarity**? For ridehail, I explored how gamification can evoke joyful passenger engagement with latent driving conditions [10]. After developing six interactive in-ride games to probe 19 drivers and 15 passengers during workshops, I uncovered desires for unforced conversational interactions, passenger knowledge gaps (*e.g.* long-term health impacts, rating-imposed service expectations), design tradeoffs around managing content (*i.e.* lighter and approachable topics are less thought-provoking) and consent – how much probing and sharing is socially appropriate in a ride?

Future Research Aims

Designing Futures of Work that Capture Labor & Evolve *Individual Human Capabilities*. Outside labor relations, workers and consumers increasingly produce value and data for platforms. For instance, the work of visual artists is often appropriated – without attribution or compensation – to produce value for AI vendors, devaluing their labor, agency and ownership while reinforcing normative biases. As AI permeates knowledge work domains, we must equip future workforces with tools that uplift and augment their distinct human capabilities. To support creative professionals, I advised a team across HCI and AI to design and evaluate POET, a tool that identifies and expands visual homogeneities of text-to-image models to **automatically generate more diverse outputs** [9] – this framework can extend to other modalities such as music and video.

Besides creativity, emergent interactive and sensing modalities such as video and speech offer exciting pathways for capturing tacit knowledge, invisible labor and latent expertise of humans. The high effort required to document hidden costs and labor often demotivates workers (especially more service-based, low-income and vulnerable workers) from actively engaging in collectivist data-sharing practices. I will explore how these alternative, richer and more effortless collection mechanisms can (1) help workers gather evidence to *visibilize and value labor*, as well as (2) *amplify human knowledge and capabilities* more broadly.

Reorienting Social Relations to Motivate Co-Liberating Collective Actions. Gamification [10] represented one technique to incentivize consumer engagement and solidarity with workers. Another way to foster unity between consumers and workers is to bring to users' attention that ***gig workers are also consumers*** of platform resources such as their client pools and matching algorithms – an opportunity for growing solidarity. What types of consumer boycotts will help workers effectively resist exploitative platform practices?

Beyond gig labor markets, union membership (currently <10% of US adults [e]) in general is declining, while labor regulations and protections are falling behind rapidly evolving technologies and worker needs. As growing segments of the workforce upskill and adapt to AI-assisted and AI-mediated workflows, I will conduct multi-stakeholder investigations to explore new ways of mobilizing and fostering solidarity among workers and their allies, so as to enable peer support and collective influence over oppressive practices.

Building Deliberative, Pluralistic and Trust-based *Communities*. Online communities of today (*e.g.* social media, OSS environments) suffer from misaligned incentives and disruption – *e.g.* inappropriate comments [4]. Sustainable communities depend on trust, requiring shared understandings of on-the-ground realities and negotiations to reconcile pluralistic values and objectives. PolicyCraft showed how human and AI-generated cases helped facilitate deliberation in education [8]. Many emerging AI capabilities can transform our trust and understanding for each other (*e.g.* persuasion [11]), and I plan to build on novel and scalable techniques to promote prosocial and deliberative community practices – starting with familiar domains of gig work and art [12], before delving into more creative and nascent realms like music, game design [10] and world-building.

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